Palliative Care HMIS Gap Analysis Report

CV Raman Health Product vs Palliative Care PRD Requirements

Version: 1.0 | **Date:** March 17, 2026 | **Prepared For:** CV Raman Health Product Team

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Executive Summary

This document provides a comprehensive gap analysis between the current CV Raman Health product and the Palliative Care Health Records & HMIS Product Requirements Document (PRD v1.0). The analysis identifies:

1. **Features Already Implemented** - Existing capabilities that meet PRD requirements
2. **Features That Can Be Adapted** - Existing features that need enhancement or modification
3. **Missing Features** - Completely new functionality required

Key Findings Summary

CATEGORY	COUNT	PERCENTAGE
✅ Already Implemented	12 modules	35%
🔄 Can Be Adapted	8 modules	24%
❌ Missing Completely	14 modules	41%

Critical Gaps

1. **Medicine Inventory Management** - Completely missing (entire module needed)
2. **Deceased Patient Records** - No workflow for end-of-life management
3. **Vital Trends Dashboard** - No graphical visualization with target lines
4. **Route Optimization** - No geolocation or traffic-aware scheduling
5. **Mobile App for Field Staff** - No offline-capable mobile application

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1. Health Records & Vital Trends

PRD Requirements

Patient Record Structure

- Demographics (name, DOB, gender, contact, emergency contact, caregiver details)
- Clinical Information (diagnoses, allergies, medications, procedures)
- Vital Signs (BP, HR, RR, temp, SpO2, weight, BMI, pain score, functional status)

Vital Trends Dashboard

- Multi-vital line graphs with overlay capability
- Time range filters (7, 14, 30, 90 days, custom)
- Selectable parameters (toggle vitals on/off)
- Target trend lines (horizontal reference lines)
- Threshold alerts (visual flagging)
- Annotation support (clinician notes)
- Export capability (PNG/PDF)

Target Trend Line Configuration

- Facility-wide or patient-specific target ranges
- Min/max thresholds per vital
- Color-coded alerts

Current Product Status

✅ **Already Implemented:** - Patient demographics with ABHA ID integration - Emergency contact and caregiver management (PatientCaregiver model) - Observation model supports all vital signs - IPD VitalsRecord for daily vitals tracking - Pain score and functional status support - Basic vital sign recording

❌ **Missing:** - Vital trends dashboard with graphical visualization - Multi-vital overlay graphs - Target trend line configuration UI - Threshold alert system for vitals - Annotation capability on vital measurements - Export vitals as PNG/PDF - Time range filtering UI for vitals - Configurable clinical target ranges

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Vital trends visualization (graphs)	❌ Missing	High	Critical
Target trend lines configuration	❌ Missing	Medium	High
Multi-vital overlay capability	❌ Missing	Medium	High
Threshold alerts	❌ Missing	Medium	High
Annotation on vitals	❌ Missing	Low	Medium
Export vitals as PNG/PDF	❌ Missing	Medium	Medium

Recommendations

🔄 **Can Be Adapted:** - Existing Observation and VitalsRecord models contain the raw data - Recharts library already in use for other analytics - Need to build new frontend components: 1. VitalTrendsChart component with Recharts 2. TargetLineConfiguration component 3. VitalsFilterPanel component 4. AnnotationEditor component - Backend: Add target_range field to Observation model - Backend: Add annotation field to VitalsRecord model

Development Estimate: 3-4 weeks

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2. Medicine Inventory Management

PRD Requirements

Medicine Record Fields

- Medicine name, generic name, brand name
- Strength/dosage, form (tablet, injection, liquid)
- Batch number and expiry date
- Current quantity, reorder quantity
- Storage location

- Supplier information, cost per unit

Par Level & Sub-Par Level System

- Par Level: Maximum standard inventory quantity
- Sub-Par Level: Minimum threshold for reorder alerts
- Alert rules:
- Above Par: Green - Optimal
- At Par: Green - Ideal
- Between Par & Sub-Par: Yellow - Caution (initiate purchase order)
- Below Sub-Par: Red - Critical (emergency reorder)
- Expired/Near Expiry (<7 days): Red - Compliance alert

Dashboard & Reporting

- Inventory Health Dashboard (color-coded)
- Critical Items Report
- Expiry Tracker
- Consumption Analytics
- Purchase Order Generation

Current Product Status

❌ **Completely Missing:** - No medicine/inventory module - No pharmacy management - No prescription tracking - No stock management system - No par/sub-par level configuration - No medicine expiry tracking - No inventory alerts - No purchase order system - No consumption analytics for medicines

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Medicine master data model	❌ Missing	High	Critical
Inventory tracking system	❌ Missing	High	Critical
Par/Sub-Par level configuration	❌ Missing	Medium	Critical
Stock alert system	❌ Missing	Medium	Critical
Expiry tracking	❌ Missing	Medium	High
Inventory dashboard	❌ Missing	Medium	High
Consumption analytics	❌ Missing	High	Medium
Purchase order workflow	❌ Missing	High	Medium
Supplier management	❌ Missing	Low	Low

Recommendations

❌ **Completely New Module Required:**

This is a major gap requiring a complete new module. Suggested implementation approach:

Phase 1: Core Inventory (Critical) 1. Create database models: - Medicine (name, generic, brand, strength, form) - InventoryItem (batch, expiry, quantity, location) - StockLevel (par_level, sub_par_level, current_qty) - StockAlert (alert_type, severity, status) 2. Build inventory management UI 3. Implement stock tracking APIs 4. Create basic inventory dashboard

Phase 2: Alerts & Monitoring (High Priority) 1. Automated alert generation (par/sub-par triggers) 2. Expiry tracking (7-day warning) 3. Critical items report 4. Email/SMS notifications

Phase 3: Analytics & Optimization (Medium Priority) 1. Consumption analytics 2. Usage trend forecasting 3. Purchase order suggestions 4. Inventory optimization recommendations

Development Estimate: 8-12 weeks (full module)

3. Patient Classification & Tracking

PRD Requirements

Patient Status Categories

1. Inpatient (IP) - Admitted to hospital/hospice
2. Outpatient (OP) - Clinic visits without admission
3. Home Care - Palliative interventions at home
4. Day Care/Respite - Temporary facility care

Census Dashboard

- Real-time census by category with sparkline trends
- Ward-level IP census with bed occupancy percentages
- Date range filtering for historical census
- Quick-access patient lists
- Patient status transition workflow with audit trail

Current Product Status

✅ **Already Implemented:** - Inpatient (IP) via IPDAdmission model - Outpatient (OP) via Encounter type=OPD - Home Care via Service type=HOME_CARE - Bed management with occupancy tracking - Encounter status transitions - Basic patient lists by type - IPDDashboard with bed statistics

❌ **Missing:** - Day Care/Respite category - Dedicated census dashboard with sparklines - Real-time census metrics card - Historical census trend analysis - Ward-level census breakdown UI - Patient status transition audit view

🔄 **Can Be Adapted:** - Existing data models support census calculation - Need UI components for census visualization - Add Day Care/Respite service type

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Day Care/Respite category	❌ Missing	Low	Medium
Census dashboard UI	❌ Missing	Medium	High
Sparkline trends	❌ Missing	Low	Medium
Ward-level census breakdown	🔄 Adaptable	Low	Medium
Status transition audit view	🔄 Adaptable	Low	Low

Recommendations

🔄 **Can Be Adapted with Minor Changes:** 1. Add "DAY_CARE_RESPITE" to Service types 2. Create CensusDashboard component with: - Real-time census cards (IP, OP, Home Care, Day Care) - Sparkline trend visualization (using Recharts) - Ward-level breakdown (use existing Bed model data) 3. Enhance existing IPDDashboard 4. Add census API endpoint: `GET /api/census/summary`

Development Estimate: 2-3 weeks

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4. End-of-Life & Deceased Records

PRD Requirements

Separate Deceased Patient View

- Dedicated interface for deceased patients
- Respectful record management

Death Record Workflow

- Date and time of death
- Location of death (home, facility, hospital)
- Certifying physician and signature

- Probable cause(s) of death (immediate vs underlying)
- Funeral/burial arrangement details
- Organ donation consent

Deceased Patient Dashboard

- View-only access to deceased records
- Filter by date range of death
- Search by patient name, ID, ward
- Death summary display
- Link to complete medical records
- Archival notes for quality assurance

Data Security & Privacy

- Secure encrypted storage
- Audit logging
- No modification of archived records
- Retention policy compliance

Current Product Status

❌ **Completely Missing:** - No deceased patient management - No death record workflow - No separate view for deceased patients - No death certificate generation - No archival system for deceased records - Patient status has no "deceased" category - No mortality review features - No bereavement documentation

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Deceased patient status	❌ Missing	Low	Critical
Death record model	❌ Missing	Medium	Critical
Death record workflow UI	❌ Missing	Medium	Critical
Deceased patient dashboard	❌ Missing	Medium	High
Death certificate generation	❌ Missing	Medium	Medium
Mortality review features	❌ Missing	Low	Medium
Archival system	❌ Missing	High	High
Bereavement documentation	❌ Missing	Low	Low

Recommendations

❌ **New Module Required:**

This is a sensitive and critical gap for palliative care. Suggested implementation:

Phase 1: Core Death Records (Critical) 1. Add database models: - DeathRecord (date, time, location, certifying_physician) - CauseOfDeath (immediate_cause, underlying_cause, ICD_code) - DeceasedPatientArchive (archived_data, retention_date) 2. Add "deceased" status to Patient model 3. Create death record entry form 4. Build deceased patient list view

Phase 2: Dashboard & Workflow (High) 1. Deceased patient dashboard with filters 2. View-only access to archived records 3. Death certificate generation (PDF) 4. Mortality review interface

Phase 3: Compliance & Archival (High) 1. Automated archival workflow 2. Retention policy configuration 3. Enhanced audit logging 4. Organ donation consent tracking

Development Estimate: 6-8 weeks

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5. Home Care Intervention Scheduling

PRD Requirements

Intervention Types

- Nursing assessment, vital sign monitoring
- Medication administration, injection/IV therapy
- Wound care, dressing changes
- Pain and symptom management
- Physical therapy, mobility assistance
- Psychological and spiritual support
- Caregiver education and counseling

Scheduling Interface

- Select patient from home care census
- Define intervention type and objectives
- Set frequency (daily, weekly, etc.)
- Assign nurse/caregiver

Route Optimization

- Integrated map view (Google Maps API / HERE Maps)
- Real-time traffic data
- Automatic route suggestion algorithm:
- Minimize travel time and distance
- Respect traffic patterns and time windows
- Prioritize urgent/critical patients
- Manual drag-and-drop route modification
- Estimated travel time between patients

Traffic-Aware Scheduling

- Current and predicted traffic conditions
- Adjusted appointment times for travel duration

- Estimated arrival time (ETA) with buffer
- Alerts if round-trip exceeds shift length
- Alternative route suggestions

Visit Management (Mobile App)

- Offline capabilities
- Geolocation-based check-in/check-out
- Record vitals and assessments
- Document interventions
- Upload photos/media (wound images)
- Sync data when connectivity available

Communication & Coordination

- In-app messaging
- SMS/push notifications for schedule changes
- Real-time location sharing
- Escalation protocol

Current Product Status

✅ **Already Implemented:** - HOME_CARE service type - Service booking workflow - Appointment scheduling system - Provider assignment - SMS reminders (AppointmentReminder model) - Patient geolocation storage (PatientLocation model)

❌ **Missing:** - Route optimization engine - Maps integration (Google Maps/HERE Maps) - Real-time traffic data integration - Geolocation check-in/check-out - Mobile app for field staff - Offline capabilities - Travel time calculation - Drag-and-drop route planning UI - Map view with patient markers - In-app messaging system - Real-time location sharing - ETA calculation - Traffic-aware scheduling logic

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Google Maps integration	❌ Missing	Medium	Critical
Route optimization algorithm	❌ Missing	High	Critical
Traffic data integration	❌ Missing	Medium	High
Map view for scheduling	❌ Missing	Medium	High
Geolocation check-in/out	❌ Missing	Medium	High
Mobile app (offline)	❌ Missing	Very High	Critical
Travel time estimation	❌ Missing	Medium	High
Drag-and-drop route editor	❌ Missing	Medium	Medium
In-app messaging	❌ Missing	High	Medium
Real-time location tracking	❌ Missing	High	Medium

Recommendations

🔄 Can Be Adapted (Major Enhancements Required):

Existing home care scheduling exists but needs significant enhancements:

Phase 1: Maps & Geolocation (Critical) 1. Integrate Google Maps JavaScript API 2. Create RouteOptimization model: - route_id, date, nurse_id - patient_sequence (ordered list) - total_distance, total_time - traffic_conditions 3. Build MapSchedulerView component: - Patient markers on map - Route polylines - Traffic layer overlay 4. Add geolocation to appointments: - check_in_location, check_in_time - check_out_location, check_out_time 5. API endpoint: `POST /api/routes/optimize`

Phase 2: Route Optimization Engine (Critical) 1. Integrate Google Maps Directions API 2. Implement route optimization algorithm: - TSP (Traveling Salesman Problem) solver - Priority weighting (urgent patients first) - Time window constraints - Traffic data consideration 3. Real-time traffic integration (Google Traffic API) 4. Travel time estimation 5. ETA calculation with buffer

Phase 3: Mobile App (Critical - Separate Project) 1. React Native or Flutter mobile app 2. Offline-first architecture (local SQLite) 3. Geolocation tracking 4. Check-in/check-out with GPS 5. Vitals recording offline 6. Photo capture (wound documentation) 7. Data sync when online 8. Push notifications

Phase 4: Communication (Medium) 1. In-app messaging (WebSocket) 2. Real-time location sharing 3. Schedule change notifications 4. Escalation alerts

Development Estimate: - Phase 1 & 2: 6-8 weeks - Phase 3 (Mobile App): 12-16 weeks (separate project) - Phase 4: 4-6 weeks

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6. User Interface & Experience

PRD Requirements

Dashboard Layout

- Real-time census metrics (IP, OP, Home Care, Day Care)
- Critical alerts (medicines below sub-par, vital deviations)
- Home care scheduled visits (today/tomorrow)
- Recent admissions/discharges
- Medicine expiry alerts
- Stock status summary

Key Page Structures

- **Patient Record Page:** Demographics header, tabs (Overview, Clinical History, Vital Trends, Medications, Care Plan, Notes)
- **Inventory Management Page:** Filter/search, table with color-coded status
- **Census Dashboard:** Summary cards, ward occupancy, patient lists
- **Home Care Scheduler:** Calendar view, map view, route optimization
- **Deceased Patient Records:** Separate view-only section

Responsive Design

- Desktop: Full feature set
- Tablet: Ward rounds, point-of-care data entry

- Mobile: Field-focused with offline sync

Accessibility

- WCAG 2.1 AA compliance
- High contrast color schemes
- Keyboard navigation
- Screen reader compatibility

Current Product Status

✅ **Already Implemented:** - Dashboard with analytics overview - Patient detail page with rich information - Form-based data entry - Responsive design (desktop/tablet/mobile web) - IPD dashboard - Appointment calendar view - Analytics dashboards

❌ **Missing:** - Medicine expiry alerts on dashboard (no inventory module) - Dedicated census dashboard - Home care scheduler with map view - Deceased patient view - Native mobile app - WCAG 2.1 AA compliance documentation - Accessibility testing

🔄 **Can Be Adapted:** - Dashboard needs enhancements for palliative care - Patient page needs vital trends tab - Need new pages for inventory, census, deceased

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Enhanced dashboard for palliative care	🔄 Adaptable	Medium	High
Vital Trends tab on patient page	❌ Missing	Medium	High
Inventory Management page	❌ Missing	High	Critical
Census Dashboard page	❌ Missing	Medium	High
Home Care Map Scheduler	❌ Missing	High	Critical
Deceased Patient Records page	❌ Missing	Medium	Critical
Native mobile app	❌ Missing	Very High	Critical
WCAG 2.1 AA compliance	❌ Missing	Medium	Medium

Recommendations

🔄 **Dashboard Enhancements:** 1. Add census summary cards to main dashboard 2. Add medicine expiry alerts section (post inventory module) 3. Add today's home care visits widget 4. Add critical patient alerts (vitals deviations)

❌ **New Pages Required:** 1. VitalTrendsDashboard - Graphical vital trends 2. InventoryManagement - Medicine stock management 3. CensusDashboard - Patient census by category 4. HomeCareMAPScheduler - Map-based route planning 5. DeceasedPatientRecords - Respectful deceased patient management

Development Estimate: - Dashboard enhancements: 1-2 weeks - New pages: 8-10 weeks (combined) - Mobile app: 12-16 weeks (separate) - Accessibility audit & fixes: 3-4 weeks

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7. Technical Architecture

PRD Requirements

Third-Party Integrations

- Maps: Google Maps API or HERE Maps
- Traffic: Real-time traffic data API
- Analytics: Chart.js, D3.js, Recharts
- SMS/Notifications: Firebase Cloud Messaging, Twilio
- File Storage: AWS S3

Data Security & Compliance

- Encryption: TLS 1.2+, AES-256
- Authentication: Multi-factor authentication (MFA), RBAC
- Audit Logging: All access/modifications logged
- HIPAA Compliance
- GDPR Compliance
- Backup: Automated daily backups
- Disaster Recovery: RTO ≤ 4 hours, RPO ≤ 1 hour

System Performance

- Page Load Time: <2 seconds
- API Response Time: <500ms (95th percentile)
- Uptime: 99.5% SLA
- Concurrent Users: 500+

Current Product Status

✅ **Already Implemented:** - Recharts for analytics - PostgreSQL on Supabase (with backups) - TLS encryption (via Supabase) - AES-256 equivalent (Fernet encryption for sensitive data) - RBAC (admin, data_entry, viewer) - Audit logging (ConsentAccessLog, AuditLog) - DPDPA 2023 compliance (India's GDPR equivalent) - JWT authentication

❌ **Missing:** - Google Maps API integration - Traffic data API integration - Multi-factor authentication (MFA) - SMS integration (Twilio) - File storage (AWS S3 or equivalent) - HIPAA compliance documentation - GDPR compliance documentation - Performance monitoring - Uptime SLA monitoring - Disaster recovery plan documentation

🔄 **Can Be Adapted:** - Firebase Cloud Messaging can be added - Twilio integration needed for SMS reminders - MFA can be added to existing auth system

Gap Analysis

FEATURE	STATUS	EFFORT	PRIORITY
Google Maps API	❌ Missing	Medium	Critical
Traffic Data API	❌ Missing	Medium	High
Twilio SMS integration	❌ Missing	Low	High
Firebase Cloud Messaging	❌ Missing	Medium	Medium
AWS S3 file storage	❌ Missing	Medium	Medium
Multi-factor authentication	❌ Missing	Medium	High
HIPAA compliance audit	❌ Missing	High	High
GDPR compliance audit	❌ Missing	High	Medium
Performance monitoring	❌ Missing	Medium	Medium
Disaster recovery plan	❌ Missing	Low	Medium

Recommendations

Phase 1: Critical Integrations 1. Google Maps JavaScript API + Directions API 2. Google Traffic API 3. Twilio for SMS (enhance AppointmentReminder) 4. MFA implementation (TOTP-based)

Phase 2: Infrastructure 1. AWS S3 for file storage (wound photos, documents) 2. Firebase Cloud Messaging for mobile push 3. Performance monitoring (New Relic or Datadog) 4. Uptime monitoring (Pingdom or UptimeRobot)

Phase 3: Compliance 1. HIPAA compliance audit and documentation 2. GDPR/DPDPA compliance review 3. Disaster recovery procedures 4. Backup restoration testing

Development Estimate: - Phase 1: 4-6 weeks - Phase 2: 4-6 weeks - Phase 3: 6-8 weeks (ongoing)

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8. Success Metrics & KPIs

PRD Requirements

Clinical Outcomes

- Reduction in medication errors: Target 50%
- Vital monitoring compliance: Target 95%
- Adverse event reporting: 100% within 24h
- Pain control: ≥80% patients with pain score ≤3
- Unplanned readmissions: <15% within 30 days
- Patient/caregiver satisfaction: ≥85%

Operational Efficiency

- Stock-out incidents: <5% annually
- Medicine wastage: <3% due to expiry
- Inventory accuracy: ≥98%
- Schedule adherence: ≥95% of planned visits
- Average travel time reduction: 20% vs baseline
- Visit duration compliance: 90% within planned duration
- Data entry time reduction: 40%
- System availability: 99.5% uptime
- User adoption: 90% active monthly users within 6 months

User Satisfaction

- Staff satisfaction: ≥4/5

- Net Promoter Score: >40
- System Usability Score (SUS): ≥75/100
- Support ticket resolution: 85% within 24h

Current Product Status

✅ **Already Implemented:** - Analytics module with dashboards - Symptom tracking - Risk assessment tracking - Form submission analytics - User activity tracking

❌ **Missing:** - Medication error tracking (no inventory module) - Vital monitoring compliance metrics - Adverse event reporting workflow - Pain control dashboards - Readmission tracking - Patient satisfaction surveys - Inventory metrics (no inventory module) - Schedule adherence tracking - Travel time metrics - Visit duration tracking - Data entry time metrics - System uptime dashboard - User adoption tracking dashboard - Support ticket system - NPS/SUS scoring

Gap Analysis

METRIC CATEGORY	STATUS	EFFORT	PRIORITY
Clinical outcome tracking	🔄 Partially implemented	Medium	High
Medication error metrics	❌ Missing (needs inventory)	High	Critical
Vital compliance tracking	❌ Missing	Medium	High
Adverse event workflow	❌ Missing	Medium	High
Pain control dashboards	❌ Missing	Low	Medium
Readmission analytics	❌ Missing	Medium	Medium
Patient satisfaction surveys	❌ Missing	Medium	High
Inventory KPIs	❌ Missing (needs inventory)	High	Critical
Schedule adherence	❌ Missing	Medium	High
Travel time analytics	❌ Missing	Medium	High
System performance metrics	❌ Missing	Medium	Medium
User adoption tracking	❌ Missing	Low	Medium

Recommendations

Phase 1: Clinical KPI Dashboard 1. Create KPIDashboard component 2. Add vital monitoring compliance calculator 3. Build adverse event reporting form 4. Create pain control trend analysis 5. Add readmission tracking

Phase 2: Operational KPI Dashboard 1. Schedule adherence calculator 2. Travel time analytics (post route optimization) 3. Visit duration tracking 4. Data entry time measurement 5. System performance monitoring

Phase 3: User Satisfaction 1. Patient satisfaction survey module 2. NPS scoring system 3. SUS assessment tool 4. Support ticket integration (Zendesk/Freshdesk)

Development Estimate: 6-8 weeks

9. Implementation Recommendations

Phased Implementation Roadmap

Phase 1: Foundation & Critical Gaps (12-16 weeks)

Priority: CRITICAL

1. **Medicine Inventory Module** (8-12 weeks)
2. Medicine master data
3. Inventory tracking
4. Par/Sub-Par level system
5. Stock alerts
6. Basic inventory dashboard
7. **Vital Trends Dashboard** (3-4 weeks)
8. Graphical vital trends
9. Target trend lines
10. Multi-vital overlay
11. Threshold alerts
12. **Deceased Patient Records** (6-8 weeks)
13. Death record workflow
14. Deceased patient dashboard
15. Archival system

Total Effort: 17-24 weeks (parallel development possible)

Phase 2: Enhanced Workflows (14-18 weeks)

Priority: HIGH

1. **Home Care Route Optimization** (6-8 weeks)
2. Google Maps integration
3. Route optimization engine
4. Traffic data integration

5. Map-based scheduler
6. **Census Dashboard** (2-3 weeks)
7. Real-time census metrics
8. Sparkline trends
9. Ward-level breakdown
10. **Clinical KPI Dashboard** (4-6 weeks)
11. Vital compliance tracking
12. Adverse event reporting
13. Pain control analytics
14. Readmission tracking
15. **Patient Satisfaction Surveys** (2-3 weeks)
16. Survey form builder
17. NPS/SUS scoring
18. Satisfaction analytics

Total Effort: 14-20 weeks

Phase 3: Mobile & Advanced Features (16-24 weeks)

Priority: MEDIUM-HIGH

1. **Mobile App for Field Staff** (12-16 weeks)
2. React Native/Flutter app
3. Offline capabilities
4. Geolocation check-in/out
5. Vitals recording
6. Photo capture
7. Data sync
8. **In-App Messaging** (4-6 weeks)
9. WebSocket messaging
10. Real-time notifications
11. Provider-nurse communication
12. **Advanced Analytics** (4-6 weeks)

13. Consumption analytics
14. Forecasting
15. Operational metrics
16. Performance dashboards

Total Effort: 20-28 weeks

Phase 4: Compliance & Optimization (8-12 weeks)

Priority: MEDIUM

1. **HIPAA Compliance Audit** (4-6 weeks)
2. Documentation
3. Security hardening
4. Audit trail enhancements
5. **WCAG 2.1 AA Accessibility** (3-4 weeks)
6. Accessibility audit
7. Remediation
8. Testing
9. **Performance Optimization** (3-4 weeks)
10. Monitoring setup
11. Query optimization
12. Caching layer

Total Effort: 10-14 weeks

Total Implementation Timeline

- **Minimum:** 59 weeks (14-15 months)
- **Maximum:** 86 weeks (20-21 months)
- **Recommended (with parallel teams):** 12-15 months

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10. Cost & Effort Estimates

Development Effort Summary

MODULE	EFFORT (WEEKS)	TEAM SIZE	PRIORITY
Medicine Inventory	8-12	2-3 devs	Critical
Vital Trends Dashboard	3-4	1-2 devs	Critical
Deceased Patient Records	6-8	2 devs	Critical
Route Optimization	6-8	2-3 devs	Critical
Census Dashboard	2-3	1 dev	High
Clinical KPI Dashboard	4-6	1-2 devs	High
Mobile App	12-16	3-4 devs	Critical
In-App Messaging	4-6	2 devs	Medium
Compliance & Accessibility	10-14	2 devs	Medium
TOTAL	**55-77 weeks**	**Variable**	**Mixed**

Third-Party Service Costs (Annual Estimates)

SERVICE	PURPOSE	ESTIMATED COST
Google Maps API	Route optimization, maps	$5,000 - $15,000
Twilio SMS	Appointment reminders	$2,000 - $5,000
Firebase	Push notifications	$1,000 - $3,000
AWS S3	File storage	$500 - $2,000
Monitoring (Datadog/New Relic)	Performance monitoring	$3,000 - $8,000
TOTAL ANNUAL		**$11,500 - $33,000**

Resource Requirements

Development Team: - 2-3 Backend Developers (Python/Flask) - 2-3 Frontend Developers (React) - 2-3 Mobile Developers (React Native/Flutter) - 1 DevOps Engineer - 1 QA Engineer - 1 UX/UI Designer - 1 Product Manager - 1 Healthcare Domain Expert

Estimated Total Cost Range: - **Minimum (lean team, sequential):** $250,000 - $350,000 - **Optimal (full team, parallel):** $400,000 - $600,000 - **Maximum (comprehensive):** $700,000 - $900,000

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11. Risk Assessment

High-Risk Items

1. **Medicine Inventory Module (Critical)**
2. **Risk:** Complex module with compliance implications
3. **Mitigation:** Engage pharmacy domain expert, phased rollout
4. **Mobile App Development (Critical)**
5. **Risk:** Offline sync complexity, platform-specific issues
6. **Mitigation:** Use proven framework (React Native), extensive testing
7. **Route Optimization Algorithm (High)**
8. **Risk:** Complex optimization logic, traffic data accuracy
9. **Mitigation:** Use Google Maps APIs, incremental improvements
10. **HIPAA Compliance (High)**
11. **Risk:** Regulatory requirements, potential penalties
12. **Mitigation:** Compliance audit early, legal review

Medium-Risk Items

1. **Data Migration for Deceased Records**
2. **Third-Party API Reliability (Maps, Traffic)**
3. **User Adoption of New Features**
4. **Performance at Scale (500+ concurrent users)**

12. Conclusion & Next Steps

Summary of Gaps

Critical Gaps (Must Have for Palliative Care HMIS): 1. Medicine Inventory Management (completely missing) 2. Deceased Patient Records (completely missing) 3. Vital Trends Dashboard with graphs (missing) 4. Route Optimization for home care (missing) 5. Mobile app for field staff (missing)

High-Priority Gaps (Important for Full Functionality): 1. Census dashboard 2. Clinical KPI tracking 3. Adverse event reporting 4. Patient satisfaction surveys 5. HIPAA compliance

Medium-Priority Gaps (Nice to Have): 1. In-app messaging 2. Advanced analytics 3. WCAG 2.1 AA accessibility 4. Performance monitoring

Strengths of Current Product

Your CV Raman Health product has excellent foundations: - ✅ Robust patient management with ABHA compliance - ✅ Flexible form system with AI builder - ✅ Strong IPD management with bed tracking - ✅ Comprehensive appointment scheduling - ✅ Good analytics framework - ✅ DPDPA 2023 compliance - ✅ Cohort management - ✅ Provider scheduling

Recommended Next Steps

1. **Immediate (Next 3 Months):**
2. Develop medicine inventory module (Phase 1)
3. Build vital trends dashboard
4. Start Google Maps integration
5. **Short-Term (3-6 Months):**
6. Complete deceased patient records
7. Implement route optimization
8. Build census dashboard
9. Develop clinical KPI dashboard
10. **Medium-Term (6-12 Months):**

11. Launch mobile app development
12. Add in-app messaging
13. Conduct HIPAA compliance audit
14. Implement advanced analytics
15. **Long-Term (12+ Months):**
16. Continuous optimization
17. Accessibility improvements
18. User feedback integration
19. Scale and performance tuning

Final Recommendation

Option 1: Full Palliative Care HMIS (15-18 months) - Implement all critical and high-priority features - Comprehensive compliance audit - Mobile app development - Estimated cost: $500,000 - $700,000

Option 2: Core Palliative Care Extension (8-10 months) - Focus on critical gaps only (inventory, deceased, vital trends, route optimization) - Web-only (no mobile app initially) - Basic compliance - Estimated cost: $300,000 - $400,000

Option 3: Incremental Enhancement (Ongoing) - Prioritize features based on user feedback - Gradual rollout over 18-24 months - Lower upfront cost, longer timeline - Estimated cost: $400,000 - $600,000

Appendix A: Feature-by-Feature Comparison Matrix

PRD FEATURE	CURRENT STATUS	CAN ADAPT?	EFFORT	PRIORITY
Patient demographics	✅ Implemented	N/A	N/A	N/A
ABHA ID integration	✅ Implemented	N/A	N/A	N/A
Vital signs recording	✅ Implemented	N/A	N/A	N/A
Vital trends graphs	❌ Missing	🔄 Yes	Medium	Critical
Target trend lines	❌ Missing	🔄 Yes	Medium	High
Medicine inventory	❌ Missing	❌ No	High	Critical
Par/Sub-Par levels	❌ Missing	❌ No	Medium	Critical
Stock alerts	❌ Missing	❌ No	Medium	Critical
Expiry tracking	❌ Missing	❌ No	Medium	High
IP/OP classification	✅ Implemented	N/A	N/A	N/A
Home Care service	✅ Implemented	🔄 Yes	Low	N/A
Day Care/Respite	❌ Missing	🔄 Yes	Low	Medium
Census dashboard	❌ Missing	🔄 Yes	Medium	High
Deceased patient records	❌ Missing	❌ No	High	Critical
Death record workflow	❌ Missing	❌ No	Medium	Critical
Route optimization	❌ Missing	🔄 Partial	High	Critical
Google Maps integration	❌ Missing	❌ No	Medium	Critical
Traffic data	❌ Missing	❌ No	Medium	High

PRD FEATURE	CURRENT STATUS	CAN ADAPT?	EFFORT	PRIORITY
Geolocation check-in	❌ Missing	🔄 Yes	Medium	High
Mobile app	❌ Missing	❌ No	Very High	Critical
In-app messaging	❌ Missing	❌ No	High	Medium
MFA authentication	❌ Missing	🔄 Yes	Medium	High
HIPAA compliance	❌ Missing	🔄 Partial	High	High
WCAG 2.1 AA	❌ Missing	🔄 Yes	Medium	Medium

Legend: - ✅ Implemented: Feature exists and meets requirements - ❌ Missing: Feature does not exist - 🔄 Can Adapt: Existing infrastructure can be modified - N/A: Not applicable

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Appendix B: API Endpoints Gap Analysis

Required PRD Endpoints vs Current Product

PRD REQUIREMENT	CURRENT ENDPOINT	STATUS
GET /api/vitals/trends	None	❌ Missing
POST /api/vitals/target-ranges	None	❌ Missing
GET /api/inventory/medicines	None	❌ Missing
POST /api/inventory/stock-alert	None	❌ Missing
GET /api/inventory/expiry-tracker	None	❌ Missing
GET /api/census/summary	None	❌ Missing
GET /api/census/ward-breakdown	GET /api/ipd/dashboard (partial)	🔄 Adaptable
POST /api/deceased/death-record	None	❌ Missing
GET /api/deceased/patients	None	❌ Missing
POST /api/routes/optimize	None	❌ Missing
GET /api/routes/traffic	None	❌ Missing
POST /api/visits/check-in	None	❌ Missing
GET /api/kpi/clinical	None	❌ Missing
GET /api/kpi/operational	None	❌ Missing

Appendix C: Database Schema Changes Required

New Tables Needed

1. **Medicine** - Medicine master data
2. **InventoryItem** - Batch-level inventory tracking
3. **StockLevel** - Par/Sub-Par configuration
4. **StockAlert** - Inventory alerts
5. **StockMovement** - Inventory transactions
6. **DeathRecord** - Death information
7. **CauseOfDeath** - Death causes
8. **DeceasedPatientArchive** - Archived records
9. **VitalTargetRange** - Target trend lines
10. **VitalAnnotation** - Clinician notes on vitals
11. **Route** - Optimized route plans
12. **RouteWaypoint** - Route patient sequence
13. **VisitCheckIn** - Geolocation check-in/out
14. **ClinicalKPI** - KPI tracking
15. **PatientSatisfactionSurvey** - Satisfaction surveys

Modifications to Existing Tables

1. **Patient** - Add deceased_status, date_of_death
2. **Service** - Add DAY_CARE_RESPITE type
3. **Appointment** - Add check_in_location, check_out_location, actual_travel_time
4. **Observation** - Add target_min, target_max, annotation
5. **VitalsRecord** - Add is_critical, alert_triggered

Appendix D: Technology Stack Additions

Required Third-Party Libraries/Services

Backend: - google-maps-services-python (Google Maps API) - python-tsp (route optimization) - celery + redis (background jobs for route calculation) - reportlab (PDF generation for death certificates) - pyotp (MFA/TOTP)

Frontend: - @react-google-maps/api (Google Maps React) - react-beautiful-dnd (drag-and-drop route editor) - recharts (already in use, expand usage) - jspdf (client-side PDF export) - socket.io-client (in-app messaging)

Mobile: - React Native or Flutter - react-native-maps (Google Maps) - react-native-geolocation - realm or watermelondb (offline storage) - react-native-camera (photo capture)

Infrastructure: - AWS S3 or Cloudinary (file storage) - Twilio (SMS) - Firebase Cloud Messaging (push notifications) - Redis (caching, background jobs) - Datadog or New Relic (monitoring)

End of Gap Analysis Report

Prepared by: Claude Code **Date:** March 17, 2026 **Version:** 1.0

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